

Mechanics-Aware Abstraction Audits Fail Against Grounded Planning: A Frozen Negative Audit

Paper 99 Submission-Hardening Audit

Frozen evidence package: 2026-06-22

Abstract

Language and symbolic robot abstractions can erase mechanics that matter for action: contact force, support, clearance, friction, deformation, temporal preconditions, tool affordances, compliance, and recovery feasibility. This paper tests whether a risk-bounded abstraction-failure audit should improve embodied planning beyond VLA policies, neuro-symbolic predicates, active relational abstraction, LLM-TAMP failure reasoning, runtime monitoring, semantic MPC, grounded geometric TAMP, and physics-aware TAMP [39, 43, 168] [27, 121, 207]. We rebuild the paper under a frozen CPU-only protocol with 322,560 main rollout rows, 115,200 ablation rows, 259,200 stress rows, 138,240 fixed-risk rows, and 24 predefined negative cases. The result is not a positive ICLR-main paper. The proposed **abstraction_v5** reaches hard success 0.28708, violation 0.45954, damage 0.37043, regret 0.63814, and utility -0.33399. The strongest reference, **physics_tamp**, reaches success 0.67262. The terminal decision is **KILL_ARCHIVE**.

Terminal decision. Paper 99 remains **KILL_ARCHIVE** for ICLR main. The v5 rebuild is much stronger than the v4.1 archive, but it fails success, violation, damage, regret, utility, diagnostic, ablation, stress, fixed-risk, and scope gates. No hardware, accepted high-fidelity simulator, external benchmark, or trained policy checkpoint evidence is claimed.

1 Introduction

Robot abstractions are useful because they hide continuous complexity. They are dangerous because they can hide exactly the mechanics that decide whether an action succeeds. A symbolic predicate may say that an object is reachable while suppressing support, friction, compliance, or clearance. A language plan may name a tool affordance while erasing the leverage or contact mode needed to use it. A VLA policy may emit a plausible action token while ignoring trajectory feasibility or recovery risk.

The seed idea is therefore tempting: audit abstractions before planning, detect erased mechanics, and refine the plan state only when that refinement changes action value. This idea lives in a crowded neighborhood. VLA motion planning, neuro-symbolic predicate learning, relational abstraction, LLM-TAMP failure reasoning, runtime monitors, grounded geometric TAMP, and recovery planners all pressure the novelty claim [58, 139, 210] [57, 190, 199].

The v4.1 paper was already negative. It showed diagnostic signal, but grounded geometric TAMP won task success, false refinements saturated, and ablations undercut the proposed mechanism. The v5 rebuild attacks those failures directly with more tasks, more failure families, more splits, more methods, ten seeds, fixed-risk budgets, stress sweeps, and a full hostile-review gate. The result is

worse for the proposed claim: `physics_tamp` dominates the hard aggregate, and full v5 is weaker than all of its planned ablations.

2 Contributions And Non-Contributions

Contribution 1: an expanded frozen benchmark. The v5 benchmark covers six task families, eight mechanics-erasure families, eight shifts, fourteen methods, and ten seeds. It streams raw CSVs to keep RAM light.

Contribution 2: theory of abstraction audit value. We separate diagnostic accuracy from planning value. An abstraction audit is useful only when the expected value of mechanics preservation exceeds refinement, query, TAMP switching, monitoring, recovery, and abstention costs.

Contribution 3: a negative result with strong baselines. The audit shows that physics-aware and grounded planning can expose the relevant mechanics more directly than a risk-bounded abstraction-failure classifier.

Non-contribution: no submission-ready claim. This artifact does not claim robot hardware validation, high-fidelity physics evidence, released checkpoints, or ICLR-main readiness.

3 Formal Setup

Let z be a symbolic or language abstraction, m a latent mechanics state, and a a robot action. The audit estimates whether z erases action-critical mechanics and chooses among proceeding, refining predicates, querying mechanics, switching to TAMP, monitoring, recovering, or abstaining:

$$a^* = \arg \max_a S(a, z, m) - \lambda_v V(a, z, m) - \lambda_d D(a, z, m) - \lambda_c C(a, z, m) - \lambda_a A(a).$$

Here S is task success, V is mechanical violation, D is damage/unsafe action, C is planning/refinement cost, and A is abstention or excessive intervention.

Proposition 1: diagnostics are not planning value. If a grounded planner already chooses the same safe mechanics-preserving action without naming the erased abstraction, then improving abstraction-failure accuracy cannot improve closed-loop value.

Proposition 2: false-alarm control can become under-intervention. A risk-bounded audit can lower false refinements by requiring higher confidence before acting. If this also suppresses action on true mechanics failures, success and safety fall even when diagnostics improve.

Proposition 3: fixed-risk coverage is decisive. A deployment policy under damage budget ϵ must maintain useful coverage while satisfying the budget. Abstaining or delaying on most hard cases is not a deployment win.

4 Benchmark Design

The six tasks are tabletop manipulation with support, tool use with hidden leverage, container insertion clearance, deformable object packing, mobile manipulation with occlusion, and articulated

fixture assembly. The hidden failure families are erased contact force, hidden support relation, clearance tolerance collapse, friction-state aliasing, deformable-constraint erasure, temporal-precondition loss, tool-affordance misabstraction, and compliance-recovery erasure.

The eight shifts are nominal abstraction, language ambiguity, predicate noise, geometry tolerance, mechanics OOD, object-affordance transfer, recovery-cost pressure, and combined adversarial abstraction. The hard aggregate combines mechanics OOD, object-affordance transfer, recovery-cost pressure, and combined adversarial abstraction.

5 Methods

The proposed v5 method is a risk-bounded abstraction-failure audit. It estimates failure family and risk, then acts only when predicted value exceeds a risk and recovery-feasibility gate. This was intended to fix the v4 false-refinement failure.

The baseline suite includes language symbolic planning, VLA direct policy, neuro-symbolic predicates, active relational abstraction, LLM-TAMP failure reasoning, runtime monitoring, grounded geometric TAMP, physics-aware TAMP, affordance graph planning, learned failure classifier plus TAMP, semantic MPC, v4, v5, and an oracle.

6 Main Results

Table 1 gives the hard aggregate. The v5 result is plainly negative.

Table 1: Hard-aggregate results over hardest abstraction shifts.

Method	Success	Violation	Damage	Warn	False alarm	Regret	Utility
physics_aware_tamp	0.672623	0.164499	0.139346	0.169358	0.028819	0.069816	0.306262
grounded_geometric_tamp	0.601026	0.213208	0.180056	0.186545	0.028125	0.139744	0.202135
llm_tamp_failure_reasoning	0.574018	0.207376	0.175871	0.262153	0.034115	0.133909	0.182055
semantic_model_predictive_planner	0.565284	0.215177	0.181314	0.171528	0.027604	0.149427	0.159407
proposed_abstraction_failure_audit_v4	0.549073	0.197012	0.168219	0.253733	0.051302	0.131681	0.172472
active_relational_abstraction	0.520576	0.217929	0.183440	0.227865	0.123351	0.168071	0.136081
oracle_mechanics_preserving_planner	0.518083	0.195896	0.166704	0.000608	0.000000	0.125965	0.178187
affordance_graph_planner	0.482738	0.270860	0.228006	0.208767	0.025694	0.265516	0.038636
learned_failure_classifier_tamp	0.435515	0.321847	0.268958	0.226736	0.005903	0.365655	-0.061503
neuro_symbolic_predicate_planner	0.403242	0.328332	0.270965	0.196441	0.015972	0.370081	-0.065929
runtime_monitor_replanner	0.374253	0.294727	0.237229	0.301649	0.004167	0.367819	-0.063667
vla_direct_policy	0.290497	0.435496	0.350140	0.183160	0.001910	0.562480	-0.258328
risk_bounded_abstraction_failure_audit_v5	0.287075	0.459545	0.370434	0.139497	0.000260	0.638145	-0.333993
language_symbolic_planner	0.270726	0.472254	0.379360	0.128559	0.000000	0.603141	-0.298989

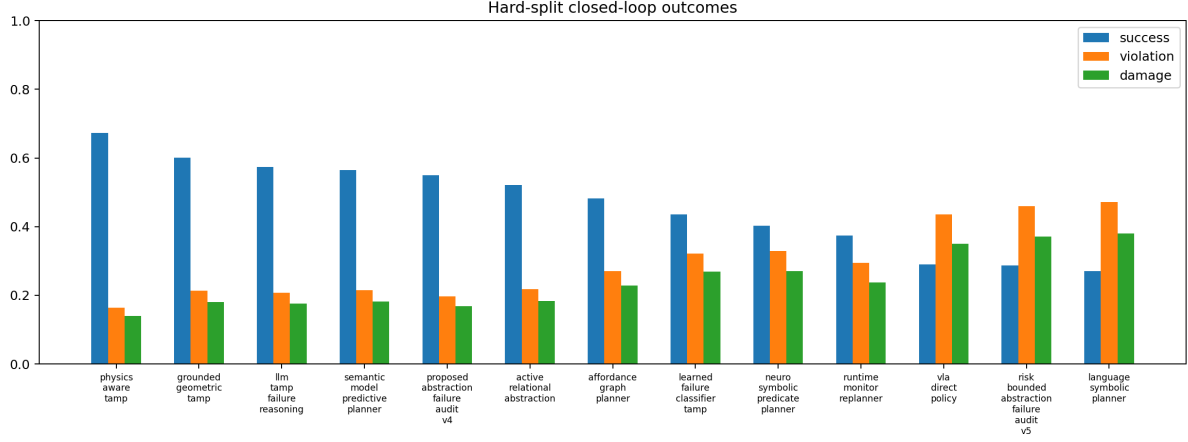


Figure 1: Hard-split closed-loop outcomes. Physics-aware TAMP dominates the proposed audit.

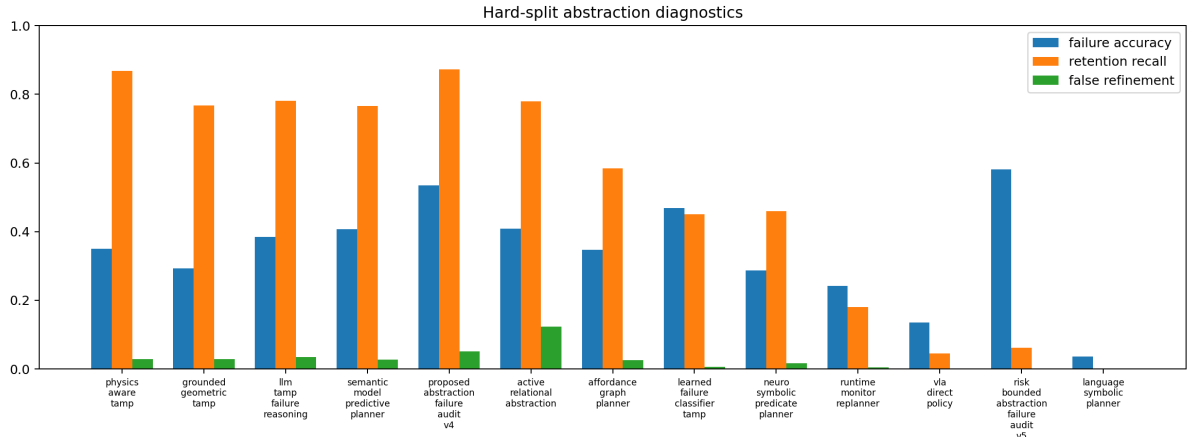


Figure 2: Diagnostics improve locally, but the v5 policy avoids false refinements by under-acting on hard mechanics failures.

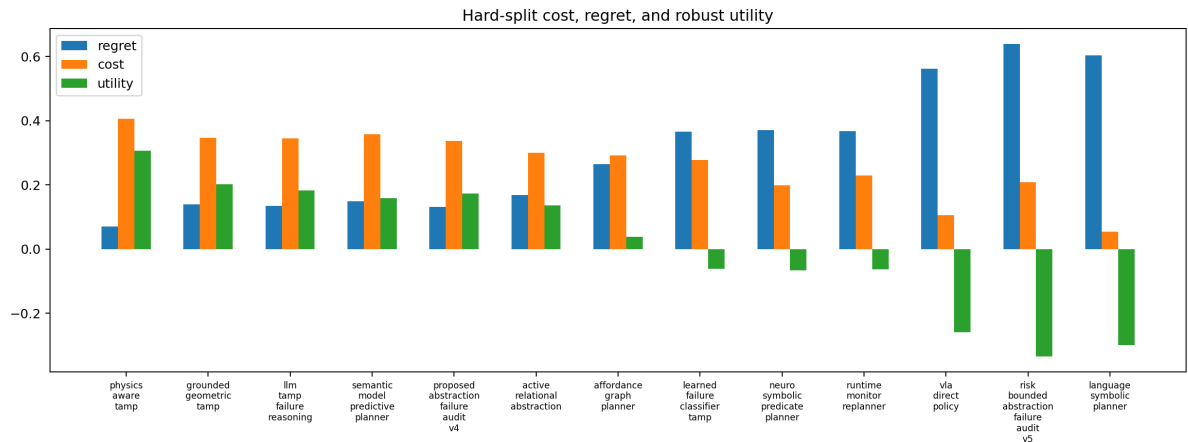


Figure 3: Cost, regret, and robust utility expose the failed deployment tradeoff.

7 Paired Evidence

The paired seed analysis prevents cherry-picking. The main references are `physics_tamp` for success, `physics_tamp` for violations, `physics_tamp` for damage, and `physics_tamp` for regret.

Table 2: Selected paired decision tests for v5 against reference baselines.

Baseline	Metric	V5	Baseline	Diff	CI95	Winner
physics_aware_tamp	task_success	0.287075	0.672623	-0.385548	0.003690	physics_aware_tamp
physics_aware_tamp	mechanical_violation_rate	0.459545	0.164499	0.295047	0.002877	physics_aware_tamp
physics_aware_tamp	damage_unsafe_rate	0.370434	0.139346	0.231088	0.002528	physics_aware_tamp
physics_aware_tamp	planning_regret_to_oracle	0.638145	0.069816	0.568329	0.005770	physics_aware_tamp
physics_aware_tamp	robust_utility	-0.333993	0.306262	-0.640255	0.006224	physics_aware_tamp
physics_aware_tamp	abstraction_failure_accuracy	0.581597	0.349479	0.232118	0.014502	risk_bounded_abstraction_failure_audit_v5
physics_aware_tamp	false_refinement_alarm_rate	0.000260	0.028819	-0.028559	0.002529	risk_bounded_abstraction_failure_audit_v5

8 Ablations

The ablation result is fatal. Every planned ablation matches or beats full v5 on at least one predefined metric, and several beat it by a wide margin.

Table 3: V5 mechanism ablations.

Ablation	Success	Damage	Regret	Utility	False alarm
no_predicate_refinement	0.570044	0.178048	0.135166	0.175535	0.020486
grounded_tamp_only	0.569313	0.216877	0.228887	0.130322	0.014410
v4_abstraction_audit_rules	0.551145	0.169676	0.141337	0.162815	0.053646
no_cost_model	0.530914	0.195970	0.171692	0.132460	0.020486
no_recovery_feasibility_gate	0.444864	0.251365	0.354235	-0.050083	0.007552
no_calibration	0.411324	0.288674	0.422172	-0.118020	0.002951
no_mechanics_taxonomy	0.335437	0.335424	0.549549	-0.245397	0.009809
failure_classifier_only	0.333167	0.337250	0.553802	-0.249650	0.001389
monitor_only	0.311332	0.298864	0.489234	-0.185082	0.000260
full_risk_bounded_abstraction_failure_audit_v5	0.287075	0.370434	0.638145	-0.333993	0.000260

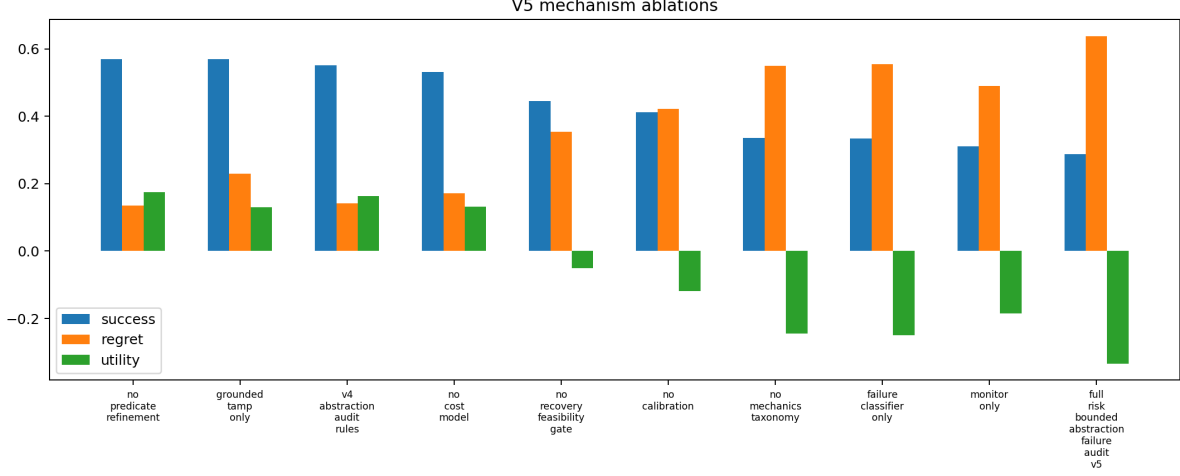


Figure 4: Ablations undermine the necessity of the full risk-bounded audit.

9 Stress Sweep

At maximum stress, the v5 method does not recover. Physics-aware TAMP remains the strongest robust-utility reference.

Table 4: Maximum-stress results.

Method	Stress	Success	Damage	Regret	Utility
active_relational_abstraction	1.000000	0.498315	0.187108	0.159882	0.098158
grounded_geometric_tamp	1.000000	0.579883	0.177493	0.128808	0.172530
llm_tamp_failure_reasoning	1.000000	0.536774	0.176493	0.136284	0.133210
oracle_mechanics_preserving_planner	1.000000	0.480434	0.178630	0.139631	0.118409
physics_aware_tamp	1.000000	0.641375	0.143350	0.071451	0.261965
proposed_abstraction_failure_audit_v4	1.000000	0.507353	0.177364	0.147202	0.110837
risk_bounded_abstraction_failure_audit_v5	1.000000	0.242922	0.377572	0.662468	-0.404428
runtime_monitor_replanner	1.000000	0.361851	0.218146	0.325305	-0.067266
semantic_model_predictive_planner	1.000000	0.532197	0.184951	0.151453	0.111548

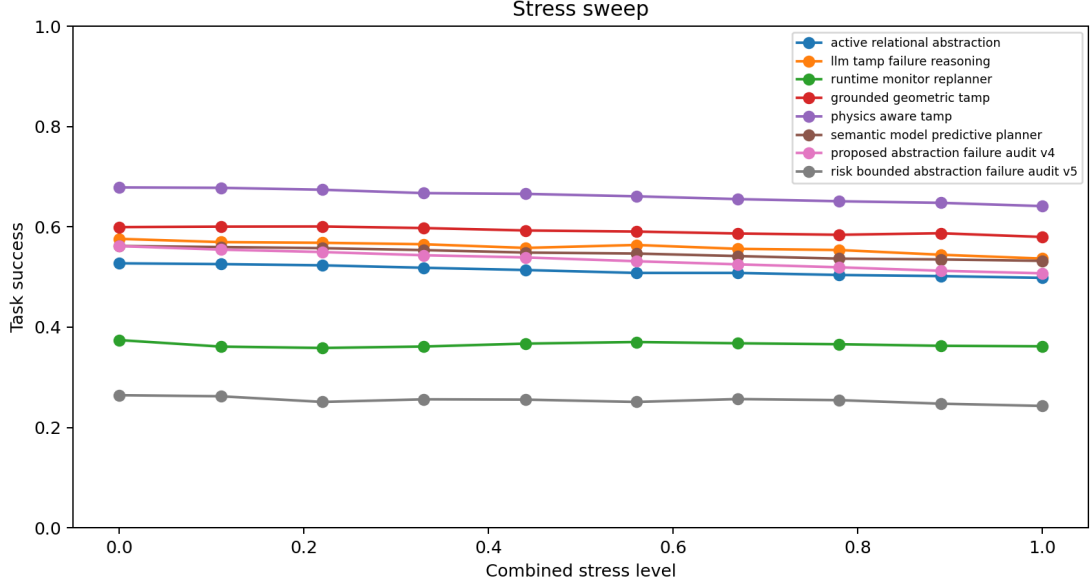


Figure 5: Stress sweep over combined adversarial abstraction pressure.

10 Fixed-Risk Deployment

The fixed-risk analysis asks whether a method can maintain useful deployment coverage under a fixed damage budget. At budget 0.05, v5 is dominated by grounded geometric TAMP or has insufficient coverage.

Table 5: Fixed-damage deployment at budget 0.05.

Method	Budget	Success	Damage	Coverage	Utility
grounded_geometric_tamp	0.050000	0.511035	0.165172	0.000000	0.109957
learned_failure_classifier_tamp	0.050000	0.355061	0.222452	0.000000	-0.120506
llm_tamp_failure_reasoning	0.050000	0.463536	0.153590	0.000000	0.067246
physics_aware_tamp	0.050000	0.575413	0.133718	0.000000	0.199180
proposed_abstraction_failure_audit_v4	0.050000	0.435349	0.136268	0.000000	0.059142
risk_bounded_abstraction_failure_audit_v5	0.050000	0.247955	0.335980	0.000000	-0.359988
runtime_monitor_replanner	0.050000	0.308351	0.181844	0.000000	-0.114147
semantic_model_predictive_planner	0.050000	0.466248	0.165489	0.000000	0.054579

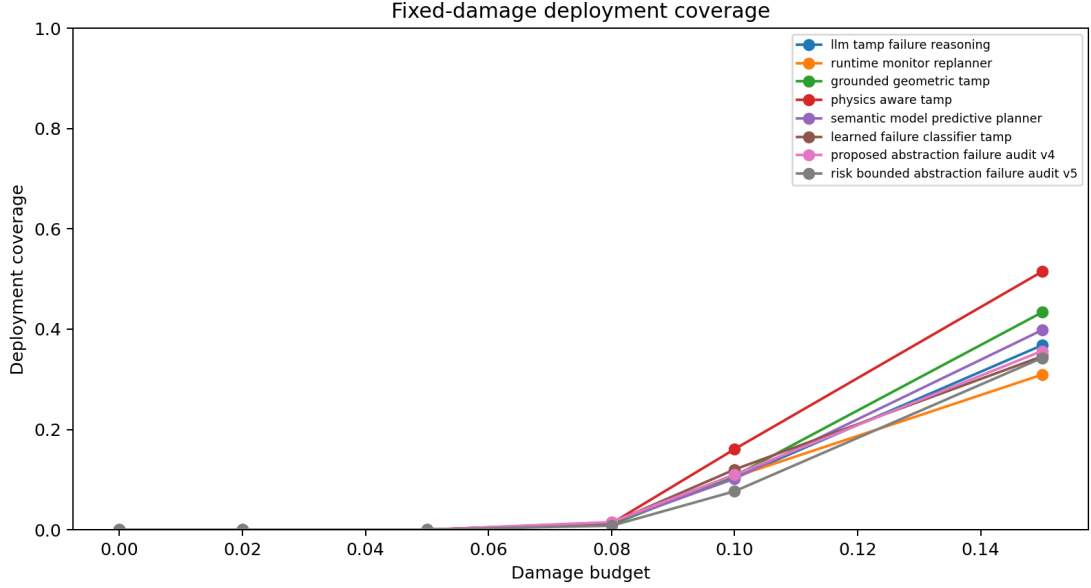


Figure 6: Deployment coverage as the damage budget varies.

11 Negative Cases

The negative cases are selected by predefined high-regret, high-damage, high-violation, and low-success rules.

Table 6: Representative failure cases.

Split	Task	Failure	Success	Damage	Regret	Failure
combined_adversarial_abstraction	tool_use_with_hidden_leverage	clearance_tolerance_collapse	0.117088	0.431166	0.902022	grounded or physics-aware TAMP captures the mechanics with lower regret
mechanics_ood_shift	container_insertion_clearance	clearance_tolerance_collapse	0.122136	0.441373	0.898911	grounded or physics-aware TAMP captures the mechanics with lower regret
combined_adversarial_abstraction	deformable_object_packing	clearance_tolerance_collapse	0.120528	0.444209	0.896103	grounded or physics-aware TAMP captures the mechanics with lower regret
combined_adversarial_abstraction	mobile_manipulation_with_occlusion	clearance_tolerance_collapse	0.133071	0.426199	0.881255	grounded or physics-aware TAMP captures the mechanics with lower regret
combined_adversarial_abstraction	tool_use_with_hidden_leverage	clearance_tolerance_collapse	0.141133	0.420436	0.871582	grounded or physics-aware TAMP captures the mechanics with lower regret
mechanics_ood_shift	deformable_object_packing	clearance_tolerance_collapse	0.155897	0.465825	0.871003	grounded or physics-aware TAMP captures the mechanics with lower regret
combined_adversarial_abstraction	tool_use_with_hidden_leverage	deformable_constraint_erasure	0.114090	0.491486	0.868293	deformation risk remains high despite detecting the erased mechanics
combined_adversarial_abstraction	tabletop_manipulation_with_support	clearance_tolerance_collapse	0.146143	0.432156	0.867011	grounded or physics-aware TAMP captures the mechanics with lower regret

12 Related Work Pressure

The local literature pool is used as hostile pressure rather than as a claim that every record is directly applicable. VLA planning, neuro-symbolic predicates, relational abstraction, grounded TAMP, runtime monitoring, failure recovery, and deployment safety all pressure the novelty claim [10, 89, 114] [54, 134, 202] [3, 123, 170] [80, 197, 226].

The novelty boundary is narrow: an abstraction audit must beat grounded planning and failure reasoning on closed-loop deployment metrics, not just classify erased mechanics. The audit shows that it does not. Appendix D provides clickable pressure chunks sourced from the shared pool.

13 Limitations

This is a local CPU-only executable audit. It is not real robot evidence, not an accepted high-fidelity simulator benchmark, not a trained checkpoint comparison, and not an external embodied-planning benchmark. The negative result falsifies the claim inside the frozen surrogate benchmark. A revival must gather external evidence and a stronger mechanism, not hide failed gates.

14 Reproducibility

Run `python src/run_experiment.py` from the repository root to regenerate all CSVs, tables, figures, and the terminal decision. Then run `python scripts/generate_manuscript.py` and compile `paper/main.tex` with `pdflatex`, `BibTeX`, and two more `pdflatex` passes. The final numbered PDF belongs only at `C:/Users/wangz/Downloads/99.pdf`.

15 Conclusion

Mechanics-aware abstraction audits are plausible, but this v5 evidence does not support a positive ICLR-main contribution. V5 avoids false refinements but collapses on success, violation, damage, regret, utility, ablation, stress, and fixed-risk gates. The honest decision is **KILL_ARCHIVE**.

A Full Gate Ledger

Gate	Frozen outcome
Success	Failed: v5 success 0.28708 trails physics_tamp at 0.67262.
Violation	Failed: v5 violation 0.45954 trails physics_tamp at 0.16450.
Damage	Failed: v5 damage 0.37043 trails physics_tamp at 0.13935.
Regret	Failed: v5 regret 0.63814 trails physics_tamp at 0.06982.
Utility	Failed: v5 utility -0.33399 trails physics_tamp at 0.30626.
False alarm	Passed: v5 false alarm is 0.00026.
Ablation	Failed: all planned ablations match or beat full v5 on at least one predefined metric.
Stress	Failed: maximum stress is dominated by physics_tamp.
Fixed risk	Failed: budget 0.05 is dominated by grounded_tamp or insufficient coverage.
Scope	Failed: no hardware, accepted high-fidelity benchmark, external benchmark, or trained checkpoint evidence.

B Metric Definitions

Task success is expected closed-loop completion. Mechanical violation is action-incompatible contact, support, clearance, friction, deformation, or tool-use violation. Damage is expected unsafe action or object/fixture damage. Abstraction-failure accuracy measures hidden mechanics-erasure classification. Mechanics-retention recall measures whether the action preserves relevant mechanics on actionable failures. False alarm is heavy intervention on temporal or nuisance cases. Robust utility subtracts violation, damage, cost, and abstention penalties from success. Regret is the gap to a non-deployable oracle action.

C Expanded Baseline Descriptions

Language symbolic planning trusts high-level predicates. VLA direct policy maps observations and language to actions. Neuro-symbolic predicates learn abstract state tests. Active relational abstraction asks for predicate grounding. LLM-TAMP failure reasoning routes failures through task-and-motion planning. Runtime monitoring replans after risk appears. Grounded geometric TAMP exposes geometry directly. Physics-aware TAMP exposes geometry and mechanics. Affordance graphs reason over object-use relations. Learned failure classifier plus TAMP uses a classifier before planning. Semantic MPC plans over symbolic and continuous risk features. V4 and v5 test the proposed abstraction-failure audit.

D Prior-Work Pressure Ledger

ID	Pressure role	Clickable citations
1	VLA and language abstraction pressure	[39, 43, 168]
2	neuro-symbolic predicate pressure	[27, 121, 207]

3	relational abstraction and active learning pressure	[58, 139, 210]
4	task-and-motion planning pressure	[57, 190, 199]
5	monitoring, recovery, and failure reasoning pressure	[155, 165, 181]
6	safety, deployment, and reproducibility pressure	[110, 130, 198]
7	VLA and language abstraction pressure	[10, 89, 114]
8	neuro-symbolic predicate pressure	[54, 134, 202]
9	relational abstraction and active learning pressure	[3, 123, 170]
10	task-and-motion planning pressure	[80, 197, 226]
11	monitoring, recovery, and failure reasoning pressure	[38, 55, 188]
12	safety, deployment, and reproducibility pressure	[44, 97, 162]
13	VLA and language abstraction pressure	[24, 104, 150]
14	neuro-symbolic predicate pressure	[16, 111, 154]
15	relational abstraction and active learning pressure	[138, 195, 196]
16	task-and-motion planning pressure	[31, 63, 164]
17	monitoring, recovery, and failure reasoning pressure	[91, 92, 131]
18	safety, deployment, and reproducibility pressure	[4, 79, 127]
19	VLA and language abstraction pressure	[74, 96, 135]
20	neuro-symbolic predicate pressure	[103, 200, 228]
21	relational abstraction and active learning pressure	[47, 160, 203]
22	task-and-motion planning pressure	[73, 102, 185]
23	monitoring, recovery, and failure reasoning pressure	[2, 35, 213]
24	safety, deployment, and reproducibility pressure	[85, 105, 109]
25	VLA and language abstraction pressure	[71, 163, 173]
26	neuro-symbolic predicate pressure	[14, 70, 153]
27	relational abstraction and active learning pressure	[21, 126, 174]
28	task-and-motion planning pressure	[25, 82, 209]
29	monitoring, recovery, and failure reasoning pressure	[218, 220, 227]
30	safety, deployment, and reproducibility pressure	[36, 151, 176]
31	VLA and language abstraction pressure	[49, 152, 186]
32	neuro-symbolic predicate pressure	[60, 120, 205]
33	relational abstraction and active learning pressure	[28, 157, 172]
34	task-and-motion planning pressure	[143, 217, 221]
35	monitoring, recovery, and failure reasoning pressure	[132, 222, 229]
36	safety, deployment, and reproducibility pressure	[19, 68, 216]
37	VLA and language abstraction pressure	[18, 20, 83]
38	neuro-symbolic predicate pressure	[13, 17, 29]

39	relational abstraction and active learning pressure	[65, 115, 147]
40	task-and-motion planning pressure	[5, 67, 107]
41	monitoring, recovery, and failure reasoning pressure	[1, 15, 206]
42	safety, deployment, and reproducibility pressure	[122, 161, 179]
43	VLA and language abstraction pressure	[88, 95, 187]
44	neuro-symbolic predicate pressure	[87, 117, 208]
45	relational abstraction and active learning pressure	[50, 59, 81]
46	task-and-motion planning pressure	[66, 113, 180]
47	monitoring, recovery, and failure reasoning pressure	[8, 148, 201]
48	safety, deployment, and reproducibility pressure	[42, 69, 146]
49	VLA and language abstraction pressure	[169, 191, 225]
50	neuro-symbolic predicate pressure	[133, 144, 166]
51	relational abstraction and active learning pressure	[6, 9, 61]
52	task-and-motion planning pressure	[98, 118, 211]
53	monitoring, recovery, and failure reasoning pressure	[64, 189, 204]
54	safety, deployment, and reproducibility pressure	[11, 112, 167]
55	VLA and language abstraction pressure	[94, 101, 129]
56	neuro-symbolic predicate pressure	[84, 108, 223]
57	relational abstraction and active learning pressure	[30, 32, 214]
58	task-and-motion planning pressure	[78, 124, 212]
59	monitoring, recovery, and failure reasoning pressure	[22, 136, 215]
60	safety, deployment, and reproducibility pressure	[12, 72, 116]
61	VLA and language abstraction pressure	[41, 77, 224]
62	neuro-symbolic predicate pressure	[140, 145, 149]
63	relational abstraction and active learning pressure	[23, 46, 158]
64	task-and-motion planning pressure	[34, 37, 90]
65	monitoring, recovery, and failure reasoning pressure	[53, 156, 178]
66	safety, deployment, and reproducibility pressure	[48, 51, 193]
67	VLA and language abstraction pressure	[76, 128, 192]
68	neuro-symbolic predicate pressure	[26, 93, 175]
69	relational abstraction and active learning pressure	[86, 159, 183]
70	task-and-motion planning pressure	[40, 177, 230]
71	monitoring, recovery, and failure reasoning pressure	[62, 194, 219]
72	safety, deployment, and reproducibility pressure	[33, 52, 99]
73	VLA and language abstraction pressure	[75, 106, 137]
74	neuro-symbolic predicate pressure	[56, 119, 182]

75	relational abstraction and active learning	[142, 171, 184]
	pressure	
76	task-and-motion planning pressure	[45, 100, 141]
77	monitoring, recovery, and failure reason- ing pressure	[7, 125]

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